Elyes Darouich

AI Engineer — Agents, RAG & LLM Systems

French citizen (EU), no work permit needed in the EU/EEA or Switzerland | Based in Bizerte, Tunisia, open to relocation
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Professional Summary

AI engineer and 2026 ESPRIT engineering graduate (Master's level). My end-of-studies project at Talan, an AI ISO-compliance auditor giving cited, clause-by-clause verdicts across 110 clauses on Llama 3.3 70B, was rated “Excellent” by the jury (18.25/20 at the defense) and kept by Talan as an internal demonstrator. I build and evaluate LLM systems end to end: multi-agent orchestration, guardrailed RAG, a 1,596-test suite and LoRA fine-tuning on Apple MLX.

Technical Skills

AI / ML: Multi-Agent Orchestration, LangGraph, RAG, LLM Guardrails & Evaluation, LoRA Fine-Tuning (Apple MLX), On-Premise LLM Deployment (Ollama), MCP, Claude Agent SDK, Vector Search & Embeddings (ChromaDB, Qdrant, sqlite-vec)

Languages: Python, TypeScript, JavaScript, Swift, Java, C++, Solidity, SQL

Backend & Data: FastAPI, NestJS, Flask, Node.js, Celery, REST, WebSockets, PostgreSQL, SQLite, MongoDB, Redis

Frontend & Mobile: React, Next.js, SwiftUI, Flutter, Tailwind CSS

DevOps & Web3: Docker, GitHub Actions CI/CD, Linux, Vercel, Zero-Knowledge (Groth16, Circom, snarkjs), Hyperledger Besu

Experience

End-of-Studies AI Engineer (PFE) — Talan Tunisia Feb 2026 – Jun 2026
Tunis, Tunisia

- Architected “SMI Virtual Auditor,” an AI ISO-compliance auditor giving cited, clause-by-clause verdicts (ISO 9001/14001/27001 and Talan’s internal Forfait standard: 110 clauses) via 3-pass RAG on Llama 3.3 70B (Groq, GitHub Models, NVIDIA NIM).
- Engineered a 7-layer guardrail system (groundedness, fidelity, contradiction, confidence capping) that suppresses hallucinations, grounding verdicts via local bge-m3 embeddings on Ollama + ChromaDB.
- Built a clean-architecture FastAPI backend (JWT RS256, RBAC, Celery, PostgreSQL/Redis/MinIO, Docker, 108 tests) and a React 18 + TypeScript SPA with streaming RAG chat and PDF export.

Software Engineering Intern — Talan Tunisia Jul 2025 – Sep 2025
Tunis, Tunisia

- Built “ProjectFlow” (Flutter + NestJS): APIs under 150 ms for 100+ concurrent users at 78% test coverage, real-time Kanban and chat over WebSockets, Gemini 2.0 Flash assistant, JWT + Google OAuth RBAC, Dockerized CI/CD.

Flutter Developer Intern — Vatech WABAG Summer 2023
Tunis, Tunisia

- Built a cross-platform leave-management app (REST API, secure auth with RBAC, Firebase push notifications).

Projects

EDITH — Local-First Personal AI Companion (macOS) *Python, Swift, Ollama, MLX, LoRA, SQLite*
Fully local, persistent AI companion: an always-on daemon wakes a 9B model only when it is needed (cold start 20.7 s, warm 0.36–0.77 s to first token, under 1 GB idle). 1,596-test suite; base-model swaps gated on measured probes (10/10 instruction following, 14/14 held-out honesty); LoRA on Apple MLX (0.06% of parameters trainable, held-out validation loss 3.89 to 2.73).

Agentic Misalignment Replication — AI Safety Research *Python, Ollama, pre-registered evaluation*
Independent replication of Anthropic’s agentic-misalignment methodology on EDITH’s 9B model, persona as the only variable: 24 runs, pre-registered scoring, every scorer flag checked against the full transcript. Reports zero harmful attempts as uninterpretable, not safe; proposes in-task comprehension as a validity gate; documents a retracted interim finding. Write-up: claude.ai/artifact/7TTKjre9CBnbJv1ziY4zhn

SentinelAI *LangGraph, FastAPI, Celery, Redis, Qdrant*
Won Talan’s three-day pre-hire bootcamp leading a randomly assigned team of 5–6: a 7-agent LangGraph platform scoring geopolitical and market risk (gold, oil, S&P 500, BTC, ETH) with Monte Carlo and GARCH models, under 30 s end to end.

ZK-Attest *Circom, snarkjs, Solidity, Hyperledger Besu, Next.js*
Privacy-preserving credential verification that proves facts without revealing the underlying data: Groth16 zero-knowledge proofs, W3C Verifiable Credentials and an ERC-5192 soulbound trust chain (GDPR by design).

Agentic Valley *Claude Agent SDK, TypeScript, VS Code API*
VS Code extension showing multi-agent orchestration as a live pixel-art office: a main agent plus 6 specialized agents, each its own Claude Agent SDK session, with every delegation animated in real time.

Education

National Engineering Degree in Computer Engineering (Master's level) — ESPRIT, Tunis 2021 – Sep 2026
Specialty: Mobile Information Systems. End-of-studies project rated “Excellent” by the jury (18.25/20 at the defense).

Baccalaureate in Computer Science 2021

Certifications & Languages

Certifications (2025): NVIDIA DLI Generative AI with Diffusion Models • NVIDIA DLI Applications of AI for Predictive Maintenance • Hashgraph Developer Course • AWS Academy Cloud Foundations (course badge)

Languages: Arabic (native) • French (native) • English (C1)